

[Manuscript title]

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Abstract

[Write the final abstract after the biological results have been interpreted and verified.]

Introduction

[Add the literature-supported biological background, study rationale, and research question. Insert only verified references in `references.bib`.]

Materials and Methods

Dataset and experimental design

Publicly available expression data were obtained from GEO accession GSE98923. The baseline analysis used a balanced subset of **54 samples** comprising Cabernet Sauvignon and Pinot noir, three developmental stages (FruitSet, Veraison, and Harvest), three years (2012-2014), and three replicates per cultivar-stage-year combination.

Expression preprocessing

The processed RPKM matrix distributed with GSE98923 was used as the expression source. The selected expression matrix contained **29971 genes** across **54 samples**. Values were transformed as $\log_2(\text{RPKM} + 1)$ before co-expression analysis.

Co-expression network analysis

Gene co-expression analysis was performed in R using CEMiTool. Pearson correlation was used with an unsigned network and signed TOM. CEMiTool filtering was enabled with `filter_pval = 0.1`, `apply_vst = FALSE`, and a minimum module size of 30.

Automatic beta selection did not return a usable value in the initial baseline analysis. For reproducibility of the established baseline, the soft-thresholding power was fixed at **beta = 10** with

`set_beta = 10` and `force_beta = FALSE`. This parameter is specific to the present sample set and preprocessing and should be re-evaluated if either changes. A random seed of 1234 was used. Software versions are provided in `logs/session_info.txt`.

Results

Co-expression modules

CEMiTool identified **11 modules** in the baseline analysis.

[Describe the biologically relevant module patterns after inspecting the module profiles, enrichment results, and hub genes. Avoid interpreting module function solely from visual appearance.]

Figures

[Select only manuscript-relevant figures from `results/figures/`. The full technical report contains the complete analysis figure set.]

Tables

[Move the principal manuscript tables here. Complete machine-readable tables are stored under `results/tables/` and indexed in the supplementary manifest.]

Discussion

[Interpret the findings in relation to grape berry development and the relevant literature. Discuss limitations, including the use of processed RPKM values, the fixed $\beta = 10$ baseline choice, the selected 54-sample baseline design, and any potential year/cultivar effects.]

Conclusions

[Write concise conclusions supported directly by the results.]

Data and code availability

The analysis uses public data from GEO accession GSE98923. The reproducible project contains the sample selection, processing scripts, exported results, software-session information, and a supplementary file manifest with MD5 checksums.

References